

随机过程作业一参考答案

习题二 3, 4, 9, 10, 14, 16

3、设 $X(t) = At + (1 - |A|)B, t \geq 0$, 这里 A 和 B 独立同分布,

$$P(A=0) = P(A=1) = P(A=-1) = \frac{1}{3}$$

(1) 写出 $\{X(t)\}$ 的所有样本函数;

(2) 计算 $P(X(1)=1), P(X(2)=1)$ 和 $P(X(1)=1, X(2)=1)$.

解:

$$X(t) = At + (1 - |A|)B, t \geq 0$$

$$P(A=0) = P(A=1) = P(A=-1) = \frac{1}{3}, P(B=0) = P(B=1) = P(B=-1) = \frac{1}{3}$$

(1)

$$A=0, B=0, X_1(t) = 0$$

$$A=0, B=1, X_2(t) = 1$$

$$A=0, B=-1, X_3(t) = -1$$

$$A=1, X_4(t) = t$$

$$A=-1, X_5(t) = -t$$

(2)

$$P(X(1)=1) = P(A=0, B=1) + P(A=1) = \frac{1}{3} \times \frac{1}{3} + \frac{1}{3} = \frac{4}{9}$$

$$P(X(2)=1) = P(A=0, B=1) = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

$$P(X(1)=1, X(2)=1) = P(A=0, B=1) = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

4、 $Z(t) = AXt + 1 - A, t \geq 0$, 这里 A 和 X 相互独立, $P(A=0) = P(A=1) = \frac{1}{2}, X \sim N(1, 1)$

(1) 计算 $P(Z(1) < 1), P(Z(2) < 2), P(Z(1) < 1, Z(2) < 2)$;

(2) 计算 $\mu_Z(t), R_Z(s, t)$.

解:

$$Z(t) = AXt + 1 - A, t \geq 0, P(A=0) = P(A=1) = \frac{1}{2}, X \sim N(1, 1)$$

(1)

$$Z_1(t) = 1, Z_2(t) = Xt$$

$$P(Z(1) < 1) = P(Z_2(1) < 1 | A=1)P(A=1) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$P(Z(2) < 2) = P(Z_2(2) < 2 | A=1)P(A=1) + P(Z_1(2) < 2 | A=0)P(A=0) = \frac{1}{2} \times P(2X < 2) + \frac{1}{2} = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} = \frac{3}{4}$$

$$\begin{aligned}
 P(Z(1) < 1, Z(2) < 2) &= P(Z_2(1) < 1, Z_2(2) < 2 \mid A = 1) P(A = 1) = P(X < 1) \times \frac{1}{2} = \frac{1}{4} \\
 (2) \quad \mu_Z(t) &= E(Z(t)) = \frac{1}{2} + \frac{1}{2}t = \frac{t+1}{2} \\
 R_Z(s, t) &= C_Z(s, t) + \mu_Z(t)\mu_Z(s) = \frac{1}{2} + st
 \end{aligned}$$

9、设 $X(t) = At + B, t \geq 0$, 这里 A 和 B 独立同分布, $E(A) = \mu, D(A) = \sigma^2 > 0$

(1) 计算 $\mu_X(t), R_X(s, t), C_X(s, t)$;

(2) 若 $A \sim N(0, 1)$, 证明 $\{X(t)\}$ 是正态过程; 并求出 $X(t), X(t) - X(s), X(t) + X(s)$ 的分布.

解:

(1)

$$\mu_X(t) = E(X(t)) = E(At + B) = \mu(t + 1)$$

$$\begin{aligned}
 R_X(s, t) &= E(X(s)X(t)) = E(A^2st + AB(s+t) + B^2) \\
 &= stE(A^2) + (s+t)E(AB) + E(B^2) \\
 &= st[D(A) + E^2(A)] + (s+t)E(A)E(B) + D(B) + E^2(B) \\
 &= \sigma^2(st + 1) + (s + 1)(t + 1)\mu^2
 \end{aligned}$$

$$C_X(s, t) = R_X(s, t) - \mu_X(s)\mu_X(t) = \sigma^2(st + 1)$$

(2)

$A \sim N(0, 1), B \sim N(0, 1), X(t) = tA + B, X(t)$ 为正态分布的线性组合

$X(t) \sim N(0, t^2 + 1)$ 为正态过程

$$X(t) - X(s) = A(s - t) \sim N(0, (s - t)^2)$$

$$X(t) + X(s) = A(s + t) + 2B \sim N(0, (s + t)^2 + 4)$$

10、设 $X_0, X_1, \dots$ 独立同分布,

$$P(X_0 = 1) = p = 1 - P(X_0 = 0), 0 < p < 1.$$

令 $Y_n = X_n + X_{n+1} + X_{n+2}$. 计算

(1) Y_n 的分布律;

(2) 在 $\{Y_0 = 2\}$ 条件下, Y_1 的条件分布律;

(3) $P(Y_0 = 1, Y_1 = 0, Y_2 = 1)$;

(4) $\{Y_n\}$ 的均值函数和自协方差函数.

解:

(1)

$$P(Y_n = i) = C_3^i p^i (1 - p)^{3-i}, i = 1, 2, 3$$

(2)

$$P(Y_1 = 1 \mid Y_0 = 2) = \frac{2}{3}(1 - p), \quad P(Y_1 = 2 \mid Y_0 = 2) = \frac{1+p}{3}, \quad P(Y_1 = 3 \mid Y_0 = 2) = \frac{p}{3}$$

(3)

$$P(Y_0 = 1, Y_1 = 0, Y_2 = 1) = P(X_0 = 1, X_1 = 0, X_2 = 0, X_3 = 0, X_4 = 1) = p^2(1 - p)^3$$

(4)

$$\mu_Y(n) = 3p$$

$$\begin{aligned} C_Y(m, n) &= \text{Cov}(Y_m, Y_n), \text{ 不难看出, } |m - n| \geq 3 \text{ 时, } \text{Cov}(Y_m, Y_n) = 0 \\ |m - n| < 3 \text{ 时, } \text{Cov}(Y_m, Y_n) &= \text{Cov}(X_m + X_{m+1} + X_{m+2}, X_n + X_{n+1} + X_{n+2}) \\ &= (3 - |m - n|) \text{Var}(X_m) = (3 - |m - n|)p(1 - p) \end{aligned}$$

$$\text{综上, } C_Y(m, n) = \begin{cases} (3 - |n - m|)p(1 - p), & |n - m| < 3, \\ 0, & |n - m| \geq 3. \end{cases}$$

14、设随机过程 $\{X(t); t \in T\}, \{Y(t); t \in T\}$ 不相关,

$$Z(t) = a(t)X(t) + b(t)Y(t) + c(t), \quad t \in T$$

这里 $a(t), b(t), c(t)$ 都是通常的函数。已知 $\mu_X(t), \mu_Y(t), C_X(s, t), C_Y(s, t)$, 求 $\mu_Z(t), C_Z(s, t)$ 。

解:

$$\begin{aligned} \mu_Z(t) &= E(a(t)X(t) + b(t)Y(t) + c(t)) \\ &= a(t)E(X(t)) + b(t)E(Y(t)) + c(t) \\ &= a(t)\mu_X(t) + b(t)\mu_Y(t) + c(t) \end{aligned}$$

$$\begin{aligned} C_Z(s, t) &= \text{Cov}(a(s)X(s) + b(s)Y(s) + c(s), a(t)X(t) + b(t)Y(t) + c(t)) \\ &= R_Z(s, t) - \mu_Z(s)\mu_Z(t) \\ &= a(s)a(t)C_X(s, t) + b(s)b(t)C_Y(s, t) \end{aligned}$$

16、设随机过程 $\{X(t); t \in (-\infty, \infty)\}, \{Y(t); t \in (-\infty, \infty)\}$ 相互独立, 已知它们的均值函数和自相关函数。令 $Z(t) = X(t)Y(t), t \in (-\infty, \infty)$, 求 $\mu_Z(t), R_Z(s, t), R_{XZ}(s, t)$ 。

解:

$$\mu_Z(t) = E(X(t)Y(t)) = E(X(t))E(Y(t)) = \mu_X(t)\mu_Y(t)$$

$$R_Z(s, t) = E(X(s)Y(s)X(t)Y(t)) = E(X(s)X(t))E(Y(s)Y(t)) = R_X(s, t)R_Y(s, t)$$

$$R_{XZ}(s, t) = E(X(s)X(t)Y(t)) = R_X(s, t)\mu_Y(t)$$